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Theory of Machines End Term Examination 2019.

ans 1 (a) A flywheel is a mechanical device specifically designed to efficiently store rotational energy when the energy supplied is more than the requirement & to supply it when there is a deficiency in the energy requirement.

Applications :- I.C engine, punching press, rivetting machine.

(b) Mechanism

Machine

→ No. of bodies assembled in a way that motion of one causes constrain & predictable motion to another

→ It is a mechanism or combination of mechanism which transmit motion by definite motion

→ Mechanism transmit & modify a motion

→ Transmit & modify available energy into desired work output

→ fundamental unit of a machine

→ combination of mechanism



Ans ① Degree of freedom,

The minimum no. of independent variable required to define the position or motion of system it is known as degree of freedom.

Redundant link are formed as those link whose presence doesn't make any difference in the mechanism.

Ans ② Coriolis Acceleration.

It is the acceleration due to the rotation of the earth, experienced by particles moving along the earth & its surface. Ocean air influence by coriolis acceleration.

$$\text{arc } DE = \text{arc } FE - \text{arc } FD$$

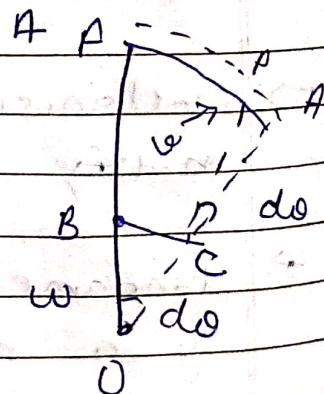
$$= (OF) d\theta - (OB) d\theta$$

$$= (OF - OB) d\theta$$

$$= CD d\theta$$

$$DE = v_w (dt)^2$$

$$DE = \frac{1}{2} f (dt)^2$$



$$\frac{1}{2} f (dt)^2 = v_w (dt)^2$$

$$f = 2v_w$$

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Ans ① Fluctuation of Energy.

The fluctuation of above & below the mean torque line in a turning moment diagram is known as coefficient of fluctuation energy.

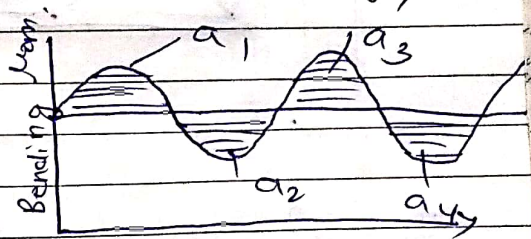
Energy at A = ϵ

$$B = \epsilon + a_1$$

$$C = \epsilon + a_1 - a_2$$

$$D = \epsilon + a_1 - a_2 + a_3$$

$$E = \epsilon + a_1 - a_2 + a_3 - a_4 \quad \text{crank angle}$$



* Fluctuation of speed.

The difference between the max & min speed during a cycle is called the max. fluctuation of speed.

The ratio of the max. deflection of speed to the mean speed is called coefficient of fluctuation of speed.

$$C_s = \frac{N_1 - N_2}{N} = \frac{\omega_1 - \omega_2}{\omega} = \frac{V_1 - V_2}{V}$$

$$N = \text{mean speed} \left(\frac{N_1 + N_2}{2} \right)$$

$$N_1 = \text{max. speed}$$

$$N_2 = \text{min. speed}$$

$$V = \text{mean speed} \left(\frac{V_1 + V_2}{2} \right)$$

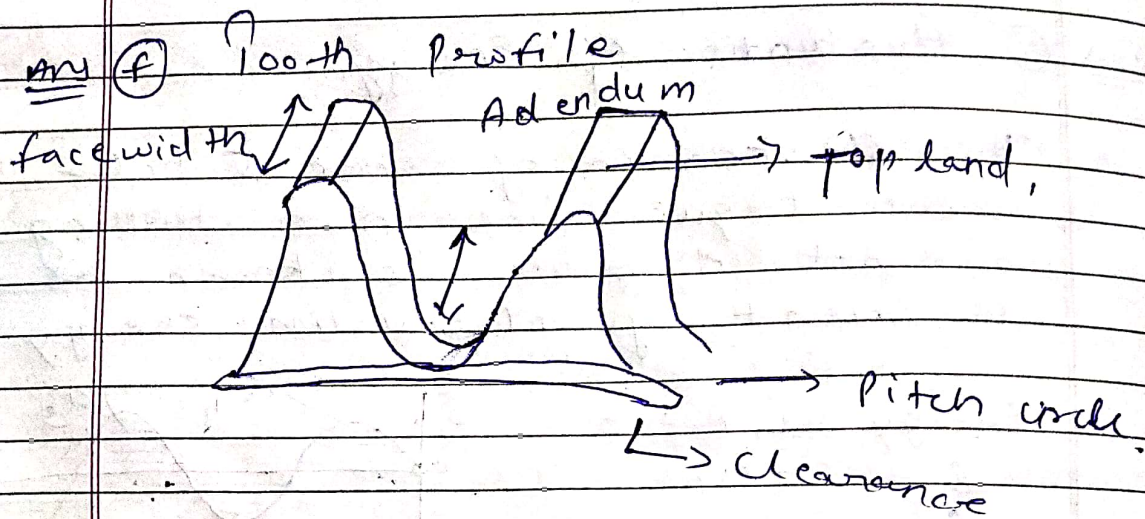
$$V_1 = \text{Max. linear speed}$$

$$\omega = \text{mean speed} \left(\frac{\omega_1 + \omega_2}{2} \right)$$

$$\omega_1 = \text{Max. angular speed}$$

$$\omega_2 = \text{Min angular speed.}$$

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Ans (g) Radial cam :-

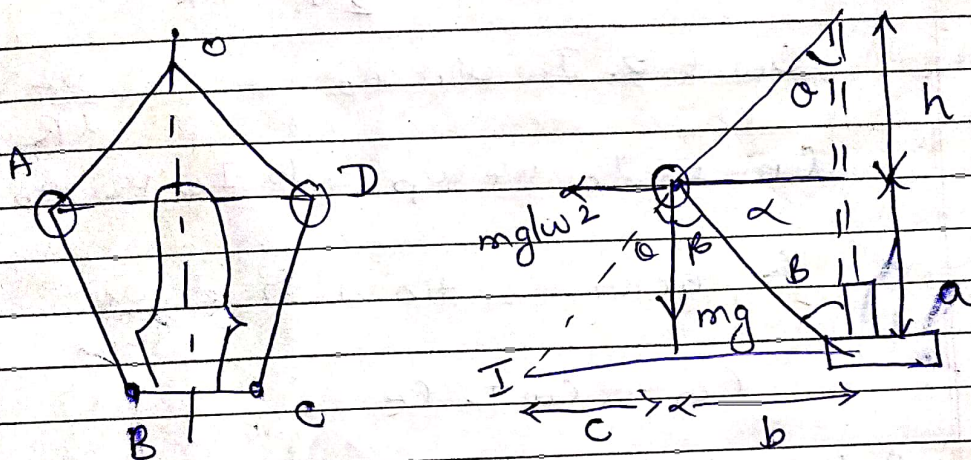
These are one of the most common cams. The radial cam is a rotating plate or disc with an outer circumference shaped to produce a required movement to a follower which is held against it.

offset cam :-

The centerline of the follower doesn't pass through the center line of the cam shaft. The amount of offset is the distance between these two center lines. The offset causes a reduction of the side thrust present in the roller follower.

Ans (h) Porter Governor :-

In the sleeve of watt governor is loaded with a heavy mass, it becomes a Porter Governor.



m = flyball mass ; M = sleeve ball

$M \gg m$ $I = I$ - centre rotation link AB

α $\rightarrow$ Upper arm vertical angle

β $\rightarrow$ Lower link

f $\rightarrow$ static friction

Considering the eqn of left hand half of the governor taking moment about I

$$mg(c) + \left(\frac{mg \pm f}{2} \right) (b+c) = (mglw^2) a$$

$$k = \frac{\tan \beta}{\tan \alpha}$$

$$N^2 = \frac{895}{h} \left[1 + \frac{(mg \pm f)(1+h)}{2mg} \right]$$

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Ans (i) Gyroscopic effect on 2 wheel.

weight on each wheel $= \frac{W}{2} = \frac{mg}{2}$ (down)

Reaction $= R_w = \frac{mg}{2}$ (up)

$$C_w = 2 \cdot I_w \omega_w \omega_p = 2 I_w \frac{V^2}{r R}$$

$$C_e = I_e \omega_e \omega_p = G I_e \omega_w \omega_p$$

$$G, \text{ gear ratio } \Rightarrow G = \frac{\omega_e}{\omega_w}$$

$$C_G = C_w + C_e$$

Force on each wheel $= C_G / 2$ (down)
Reaction on each wheel $= C_G / 2$ (up)

Ans (i) Vibration :- Any motion which repeats itself after an interval of time is called vibration or oscillation.

Reasons :-

1) Unbalanced centrifugal force caused because of non-uniform material distribution in a rotating m/c element.

2) Elastic nature of system

3) External excitation applied on the system,

- (4) winds may cause vibrations of certain systems such as electricity lines, telephone lines etc.

Unit - 1

Ans 2 (a) White worth Quick Return Mechanism
 It is a mechanism used in workshop to cut the metals. The forward stroke takes a lathe longer & cuts the metal whereas the return stroke is idle & takes a shorter period. Slider rotates in a circle about A & slider on the link. C is a pt on the link 1 extended which cuts the metal. The axis of motion of the slider passes through O & is $\perp$ to OP, fixed link.

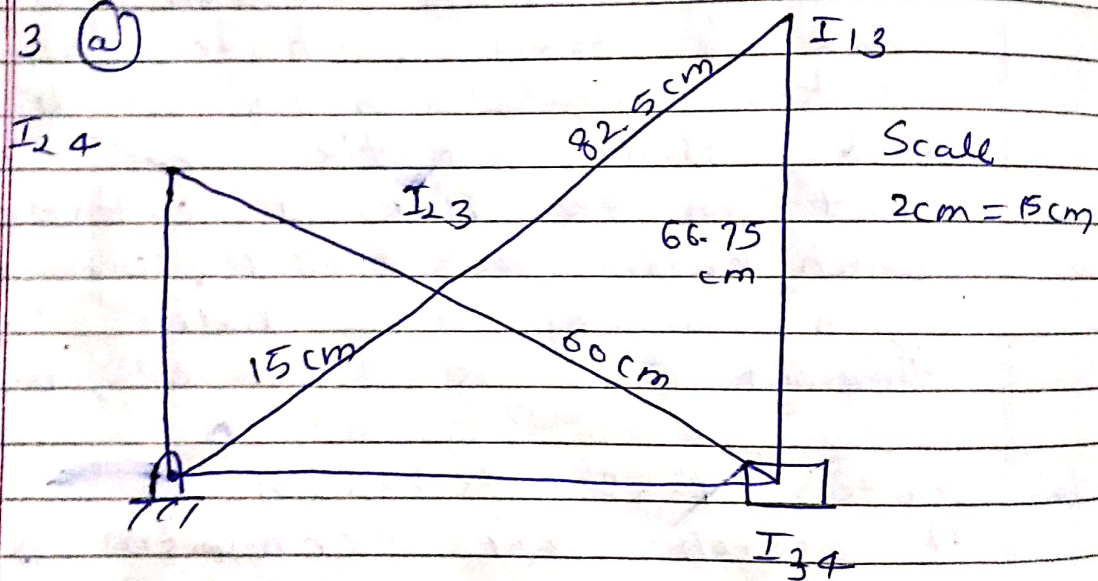
(b) Scotch Yoke Mechanism:-
 A scotch yoke mechanism is used to convert the rotary motion into a sliding motion. As the crank - 3 rotates the horizontal portion of the link 1 slides or reciprocates in the fixed link 4.

(c) Rotary Engine Mechanism:-
 It is the mechanism in which the rotation of the link 3, & the link



1 rotates about O & the slider y reciprocates on it. This also implies that of the slider is made to reciprocate on the link 1 & crank 3 will rotate about A & the link 1 about O. In rotary engine, the slide is replaced by a piston & the link 1 by a cylinder pivoted by O.

Ans 3 (a)



Given $N = 300 \text{ rpm}$

$$\Rightarrow \omega = \frac{2\pi \times 300}{60} = 31.4 \text{ rad/s}$$

$$v_B = \frac{3.14 \times 15}{100} = 4.7 \text{ m/s}$$

① velocity of slider A

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$$\omega_A = \omega_B$$

$$\frac{V_A}{I_{BA}} = \frac{V_B}{I_{BA}}$$

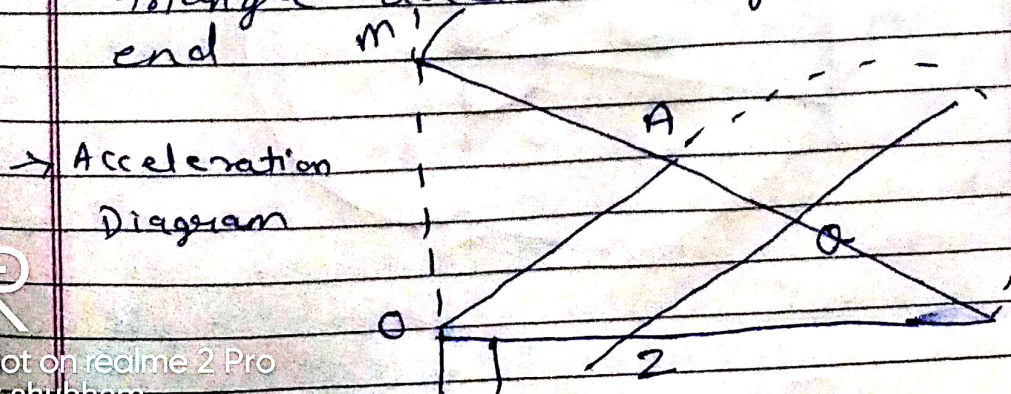
$$V_A = V_B \times \frac{I_{BA}}{I_{AA}} = 4.71 \times \frac{66.75}{82.5} = 3.81 \text{ m/s}$$

$$\textcircled{2} \quad \frac{V_A}{I_{RA}} = \frac{V_B}{I_{BB}} = w_{AB}$$

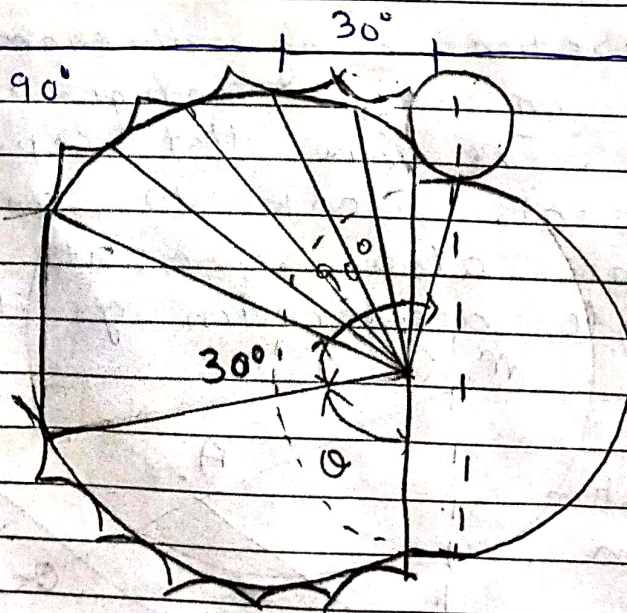
$$w_{AB} = \frac{4.71}{82.5 \times 10^{-2}} = 5.70929015$$

② KLEIN CONSTRUCTION

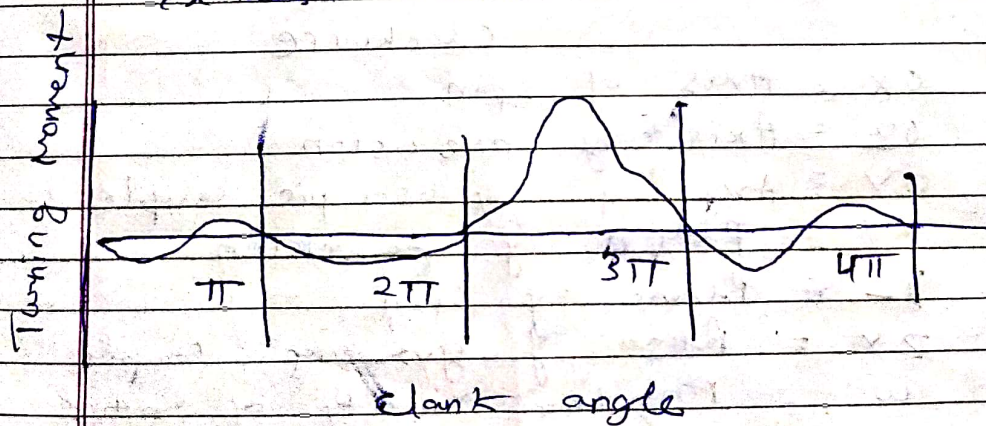
In Klein construction the velocity of acceleration along diagram are made on the configuration diagram itself. The line that separate represents crank in configuration diagram also represents velocity triangle acceleration of it's moving end m'



Ans (a) Given $\rightarrow \alpha_o = 90^\circ, \alpha_k = 90^\circ = 1.5717 \text{ rad}$
 offset = 10 mm, $S = 50 \text{ mm}$



Ans 5 a) In case of a flow stroke internal combustion engine, the diagram repeats itself after every two revolution instead of one revolution as for a steam engine. It can be seen from the diagram that for the majority of the suction stroke the turning moment is negative but becomes positive after the point P. During the compression stroke, it is totally negative & it is positive throughout the expansion stroke & again -ve for most of the exhaust stroke.

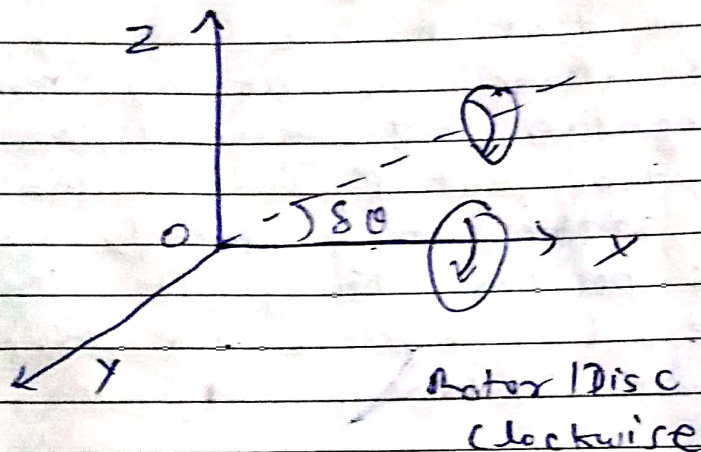


Suction compression Expansion Exhaust

b) Gyroscopic law :-

A device consisting of a wheel or disc mounted so that it can spin about an axis which is itself free to alter in direction.

It is used to provide stability or maintain a reference direction in navigation system of aircrafts, missiles, ships etc.



Ox = Axis of spin

Oz = Axis of precession

Oy = Axis of gyroscopic couple

xy = plane of precession

yz = plane of spin

zx = plane of gyroscopic couple

ω = Angular velocity of rotor

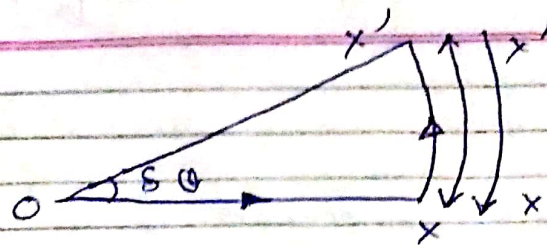
$\frac{d\theta}{dt} = \omega_p = \text{Angular velocity of precession}$

$$f = m\alpha$$

$$T = I\alpha = I\omega_p \quad (\text{magnitude})$$

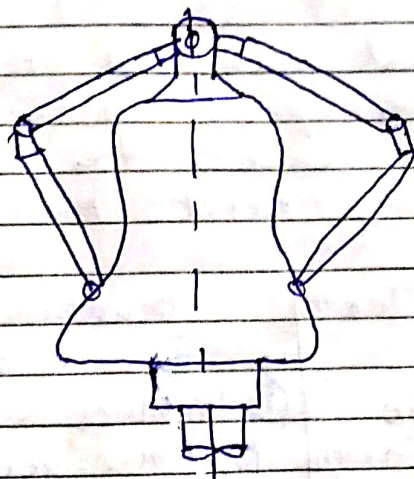
(Reactive Gyroscopic Torque)

R

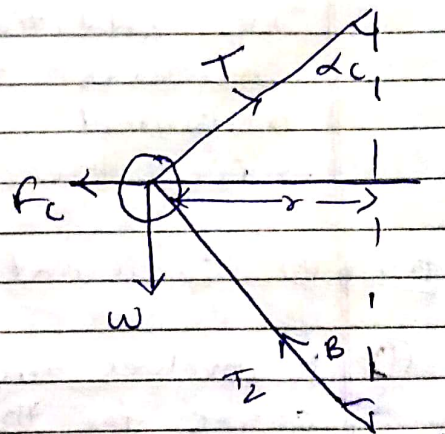


$$\alpha = \omega \frac{d\theta}{dt} = \omega \omega_p$$

Ans (6)



Sleeve



(Porter Governor)

Construction:-

The balls are connected to the arms of porter governor as shown in figure 6 these gets connected at a single point called pivot

The sleeve present in the bottom of the spindle connected to the balls to move in up or down direction, due to this momentum supply of such is done

Working :-

The porter governor is a modification of a watts governor, with central load attached to the sleeve as shown in figure. The load moves up & down the central spindle.

The additional downward forces increases the speed of revolution required to enable the ball to rise the any level predetermined.

Ans (b) Cycloidal Teeth Involute teeth

① It involves double curve for the teeth epicycloid & hypocycloid. This complete the manufacture

① Involves single curve for the teeth resulting in simple, easy of manufacturing of tools

② Difficulty in mfg.

② Simple to manufacture

③ Costlier as above result

③ cheap as above result

④ Exact centre diff. required to measure velocity ratio

④ little variation in centre differences is allowed.

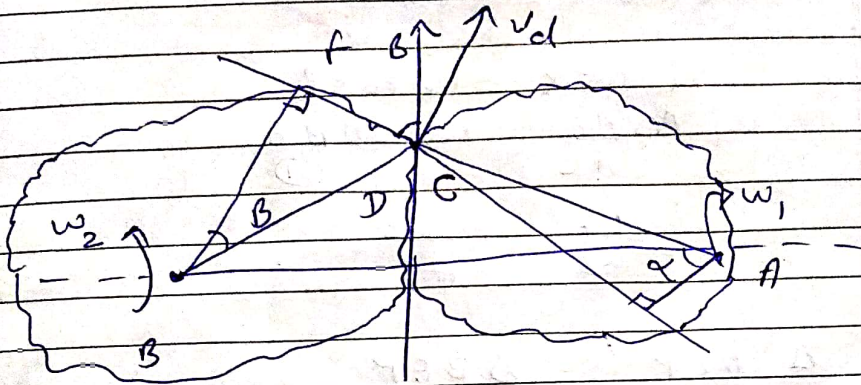
⑤ Teeths are strong

⑤ Teeths are weaker

⑥ Less wear

⑥ More wear

Ans 7 @ Law of Gearing



It states the condition which must be fulfilled by the gear tooth profile to maintain a constant angular velocity ratio between the two gears

- C → tooth profile of gear 1
- D → tooth profile of gear 2
- n-n → common normal at pt. of contact
- ω_1 → Instant A.V of gear 1
- ω_2 → Instant A.V of gear 2
- v_c → L.V of C
- v_d → L.V of D

$$v_c = \omega_1 AC$$

$$v_d = \omega_2 BD$$

To avoid the separation / penetration of the teeth, relative motion b/w the surface the common normal nn

must be zero

$$v_c \cos \alpha - v_d \cos \beta = 0$$

$$v_c \cos \alpha = v_d \cos \beta$$

$$w_1 \frac{AC \cdot AE}{AC} = w_2 \frac{BD \cdot BE}{BD}$$

$$\therefore \frac{w_1}{w_2} = \frac{BE}{AE}$$

$$\Delta AEP \sim \Delta BEP$$

$$\therefore \frac{BF}{AE} = \frac{BP}{AP} = \frac{EP}{EP}$$

$\boxed{\frac{w_1}{w_2} = \frac{BP}{AP}}$ $\Rightarrow$ Line AB is divided at P by the common normal in the inverse ratio of any velocity of two gears.

Ans ⑥ $m = 4 \text{ tonnes} = 4000 \text{ kg}$
 $k = 2 \text{ m}$

$$N_1 = 420 \text{ rpm}, \quad \omega_1 = \frac{2\pi \times 420}{60} = 43.96$$

$$N_2 = 462 \text{ rpm}, \quad \omega_2 = \frac{2 \times \pi \times 462}{60} = 48.356$$

$$I = mk^2 = 4000 \times 2^2 = 16000$$

$$\omega_{qv} = \frac{\omega_1 + \omega_2}{2} = \frac{43.96 + 48.356}{2}$$

$$w_{av} = 461.58$$

$$\begin{aligned} E &= \frac{1}{2} \times T \times w_{av}^2 \\ &= \frac{1}{2} \times \frac{8000}{16000} \times (461.58)^2 \\ &= 17,044,487.712 \text{ Nm} \end{aligned}$$

Ans $\Rightarrow M_A = 200 \text{ kg}, M_B = 300 \text{ kg}, M_C = 240 \text{ kg}$
 $M_D = 260 \text{ kg}$
 $l_A = 0.25 \text{ m}, l_B = 0.2 \text{ m}, l_C = 0.35 \text{ m}$
 $l_D = 0.3 \text{ m}$
 $\theta_A = 0, \theta_B = 45, \theta_C = 120, \theta_D = 255$

Let M_K will be the balancing mass required revolving at radius l_R & $\theta_R = 0.28 \text{ m}$
 then,

$$\begin{aligned} \Sigma F_x &= M_A l_A \cos \theta_A + M_B l_B \cos \theta_B + M_C l_C \cos \theta_C + M_D l_D \cos \theta_D \\ &= 200 \times 0.25 \cos 0^\circ + 300 \times 0.2 \cos 45^\circ + 240 \times 0.35 \cos 120^\circ + 260 \times 0.3 \cos 255^\circ \end{aligned}$$

$$\Sigma F_x = 30.23$$

$$\Sigma F_y = M_A l_A \sin \theta_A + M_B l_B \sin \theta_B + M_C l_C \sin \theta_C + M_D l_D \sin \theta_D$$

R

$$= 200 \times 0.25 \sin 0 + 300 \times 0.2 \sin 45 + 240 \times 0.3 \sin 90 + 260 \times 0.3 \times \sin 255^\circ$$

$$\Sigma F_y = 39.83$$

$$M_{xR} = \sqrt{(\Sigma F_x)^2 + (\Sigma F_y)^2}$$

$$= \sqrt{(30.23)^2 + (39.83)^2} = 50.002$$

$$M_R = \frac{50.002}{0.28} = 178.5 \text{ kg}$$

$$\tan \theta_R = \frac{39.83}{30.23} = 1.31$$

$$\theta_R = 52.64$$

my ① $m = 25 \text{ kg}$ $D = 15 \text{ kN/m}$
 $= 15000 \text{ N/m}$
 $C = 0.15 \text{ kg}$

① Critical Damping Coefficient

$$C_c = 2\sqrt{D \cdot m}$$

$$= 2\sqrt{15000 \times 25}$$

$$C_c = 1224.7 \text{ N/ms}$$

R

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② Damping factor

$$\delta = \frac{C}{C_c} = \frac{0.15 C_c}{C_c} = 0.15$$

$$\delta = 0.15$$

③ $\omega_d = \sqrt{1 - \delta^2} \omega_n$

$$\omega_n = \sqrt{\frac{D}{m}} = \sqrt{\frac{15000}{25}} = 24.5 \text{ rad/s}$$

$$\omega_d = \sqrt{1 - 0.15^2} \times 24.5$$

$$\omega_d = 24.22 \text{ rad/s}$$

④ $\Delta = \frac{2\pi\delta}{\sqrt{1-\delta^2}} = \frac{2\pi \times 0.15}{\sqrt{1-0.15^2}} = \frac{0.942}{0.988}$

$$\Delta = 0.953 \text{ Logarithmic decrement}$$

⑤ $x_n = e^{\Delta} = e^{0.953} = 2.593$
 α_{am}

Ratio of amplitude